

# Unifying Speech and Language Through Continuous Dual-Fusion for Medium-Resource Languages: The Cases of Hungarian and Greek

Anonymous Authors  
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**Abstract**—In medium-resource languages like Hungarian and Greek, Automatic Speech Recognition (ASR) suffers from scarce paired acoustic data, whereas abundant text data can be leveraged via Large Language Model (LLM) priors. To compensate for this acoustic deficit, we propose an advanced Dual-Fusion architecture that shifts ASR from isolated acoustic decoding to operating entirely within an intermediate audio-conditioned text space. Our model bridges a frozen acoustic encoder (Whisper) with pre-trained LLMs via both prompt-level embeddings and deep, continuous-space alignment across multiple intermediate decoder layers. However, unconstrained LLM priors often over-power limited acoustic signals, causing the model to over-correct grammar and drop dysfluencies. To actively control this semantic-acoustic trade-off, our methodology introduces asymmetric down-sampling to preserve fine phonetic boundaries, multi-layer gated cross-attention, target text perturbation, and a strict two-stage LoRA curriculum. Evaluated on diverse Hungarian and Greek datasets, our framework significantly reduces Word Error Rate (WER) compared to standalone decoders to both standalone acoustic decoders and acoustic-linguistic models like Vox-KriKri, demonstrating generalized performance improvements across both languages.

**Index Terms**—Automatic Speech Recognition, Speech LLMs, Continuous Fusion, Medium-Resource ASR, Multimodal Alignment

## I. INTRODUCTION

Automatic Speech Recognition (ASR) has evolved significantly with the advent of robust end-to-end models like Whisper [1] and highly data-efficient architectures like Canary [2], which achieve state-of-the-art multilingual performance. However, empirical evaluations reveal that these systems frequently exhibit performance degradation when applied to medium-resource and morphologically complex languages, such as Hungarian and Greek. This drop in accuracy is rarely due to a failure in acoustic feature extraction; rather, it manifests as orthographic and morphological errors stemming from a lack of deep, language-specific linguistic priors [3].

To bridge this gap, recent research has shifted toward multimodal Speech-Language Models (SLMs) that couple pre-trained acoustic encoders with Large Language Models (LLMs) [4]. Early integration strategies sought to minimize architectural complexity. For instance, the SLAM-ASR framework [5] demonstrated that a simple, trainable linear projector connecting an encoder like Whisper to an LLM could yield highly competitive ASR capabilities. Expanding toward broader auditory comprehension, models like SALMONN [6]

utilized Q-Formers to compress and bridge audio embeddings into the LLM space. However, rigid temporal pooling risks destroying the micro-phonetic boundaries required for highly inflected languages [7]. To achieve deeper semantic integration, researchers explored cross-attention fusion. The Whispering LLaMA framework [8] pioneered cross-modal generative error correction by integrating Whisper’s audio embeddings directly into the LLM via cross-attention adapters. Comparative studies on connector structures confirm that while multi-head cross-attention introduces more parameter overhead than simple linear layers, it provides significantly superior representational alignment [9]. Furthermore, foundational vision-language research, such as the Flamingo model [10], established the efficacy of gated cross-attention to interleave frozen text blocks with continuous multimodal inputs, ensuring stability during generation. In the speech domain, the recent VoxKriKri [11] framework operates fully within a continuous latent space, fusing Whisper’s hidden decoder states with an LLM’s textual representations at designated layers, proving that intermediate layer injection yields robust cross-modal alignment for languages like Greek.

A persistent vulnerability in SLMs, however, is the “lazy LLM” hallucination problem, where the decoder ignores the acoustic cross-attention vectors and defaults to its massive pre-trained language priors. Recent studies have shown that manipulating the text space through text perturbation and target masking forces the generative model to rely more heavily on the continuous multimodal evidence, thereby calibrating its hallucination boundaries and preventing language prior dominance.

Building upon these foundational advancements, this paper presents a novel acoustic-linguistic fusion methodology explicitly optimized for medium-resource languages, utilizing Hungarian (via *elte-nlp/Racka-4B*) [12] and Greek (via *Llama-KriKri*) [13] as primary test cases. We bypass the limitations of monolithic projection by proposing a Dual Fusion Strategy that feeds acoustic embeddings simultaneously at the prompt level and through cross-attention, utilizing asymmetric down-sampling factors to preserve phonetic granularity. To ensure deep semantic alignment without grammatical collapse, our architecture introduces Multiple Injection Layers with Layer-Wise Fusion, alongside a Gated Cross-Attention mechanism to shield syntactic generation from acoustic noise. The entire

pipeline is stabilized via a strict 2-stage LoRA curriculum and is further regulated using Text Perturbation to enforce acoustic reliance during generation.

Evaluated extensively on diverse Hungarian (utilizing the elte-nlp/Racka-4B backbone) and Greek datasets (utilizing LLAMA-KriKri-8B), our continuous dual-fusion framework significantly reduces WER across diverse Hungarian and Greek datasets, outperforming both standalone acoustic decoders and acoustic-linguistic baselines.

## II. PROPOSED DUAL-FUSION ARCHITECTURE

Our architecture leverages the combined strengths of a frozen Whisper acoustic encoder and a pre-trained LLM backbone[cite: 10].

### A. Decoupled Downsampling

To prevent the loss of critical phonetic cues, such as rapid Greek vowel transitions or brief Hungarian suffixes, we decouple the downsampling rates[cite: 10, 11]. We utilize a Simple Reshaping adapter at the prompt-embedding level for fast generation initialization[cite: 10, 11]. For the intermediate cross-attention modules, we deploy a Continuous Integrate-and-Fire (CIF) downsampler[cite: 11]. The CIF module dynamically predicts soft weights  $\alpha_i$  via a 1D convolution and a sigmoid activation:

$$\alpha_i = \sigma(W_{proj} \cdot \text{Conv1D}(X_{audio})) \quad (1)$$

These weights are integrated temporally to generate dynamically fired acoustic tokens based on acoustic complexity[cite: 10].

### B. Multiple Injection Layers and Gated Fusion

Rather than performing cross-attention at a single deep layer, we distribute the acoustic representations across multiple intermediate layers (e.g., layers 15, 25, and 35) to create a persistent acoustic memory[cite: 10]. To prevent the injected acoustic features from overwhelming the LLM’s internal linguistic representations, we introduce a gated cross-attention mechanism[cite: 10, 11]. The projected acoustic context  $C_{audio}$  is scaled by a learnable gate parameter  $g$ :

$$H_{fused} = H_{LLM} + \tanh(g) \odot C_{audio} \quad (2)$$

When generating purely syntactic tokens, the gate closes ( $g \rightarrow 0$ ), protecting the LLM’s native generative flow[cite: 10, 11].

### C. Auxiliary Multi-Task Optimization

The network is optimized using a multi-task learning objective[cite: 10, 11]:

- **Character-Level CTC Loss:** Applied directly to downsampled acoustic features to force the retention of exact phonetic information[cite: 10, 11].
- **Audio Forecasting Loss:** Predicts future acoustic frames using an MSE loss to enforce temporal reasoning[cite: 10, 11].
- **Duration Modeling:** Predicts audio snippet length to align generation length with physical speech time[cite: 11].

## III. EXPERIMENTAL SETUP

### A. Two-Stage LoRA Curriculum

Training is split into a strict two-stage curriculum[cite: 10]. In Stage 1, the base LLM and Whisper encoder are fully frozen, and only the bridge network (adapters and cross-attention modules) is trained[cite: 10, 11]. In Stage 2, Low-Rank Adaptation (LoRA) is activated to fine-tune the network end-to-end[cite: 10, 11].

### B. Regularization and Epistemic Prompting

To prevent the LLM from ignoring acoustic features, we apply text-modality dropout and blank audio training[cite: 10, 11]. Furthermore, we utilize adversarial epistemic framing by explicitly instructing the model to suppress its grammar engine using tags such as [VERBATIM] and [NORMALIZED] [cite: 11].

### C. Corpora

The models were exposed to diverse acoustics and vocabularies across a compiled corpus that includes Common Voice, FLEURS, HParl, TEDx, Logotypographia, Speech Massive, Vox Populi, Yodas, DataOcean, and Datatang[cite: 10, 11].

## IV. RESULTS AND DISCUSSION

### A. Quantitative Results

TABLE I  
WER COMPARISON ACROSS BASELINE AND PROPOSED MODELS

| Dataset      | Whisper-v3 | Ours (Greek) | Ours (Hu) |
|--------------|------------|--------------|-----------|
| Common Voice | 13.98      | 7.3          | 9.0       |
| FLEURS       | 11.02      | 10.9         | 10.2      |
| Vox Populi   | -          | -            | 9.8       |
| HParl        | 16.99      | 8.2          | -         |

Note:

Values are approximations based on internal reports.

### B. Qualitative Error Analysis

Our continuous-fusion method successfully teaches the model how sentences are structured, but occasionally at the cost of strict acoustic adherence[cite: 10]. The most frequent substitution errors stem from grammar and formatting disagreements rather than acoustic failures[cite: 10]. For example, the model consistently splits Hungarian compound words and maps acoustic sounds to Hungarian phonetic conventions (e.g., rendering "Mary" as "Méri") [cite: 10]. In cut-off audio, the LLM attempts to resolve the incomplete thought gracefully rather than stopping transcription[cite: 10].

## V. CONCLUSION

The proposed Dual-Fusion architecture successfully bridges the modality gap between acoustic encoders and text-based LLMs for Greek and Hungarian ASR[cite: 10]. While decoupled downsampling and gated cross-attention maintain high accuracy, future work will focus on integrating synthetic datasets to further mitigate the LLM "over-intelligence" paradox[cite: 10, 11].

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